Venkata Lakshmi Parimala Pasupuleti

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PROFESSIONAL SUMMARY

AI systems researcher and engineer with 8+ years of production engineering depth. Built **Leasel** -- a LangGraph-orchestrated agentic platform for Affordable Housing document analysis and ticket investigation, with a hard-coded read-only safety boundary (test-verified: never auto-resolves regulated decisions regardless of confidence), TF-IDF retrieval over incident history, and a live HUD User API integration for real-time income-limit lookups. Conducted research at George Mason University building multi-source prediction models ($R^2 = 0.977$ across 190+ countries) and applying Isolation Forest for automated anomaly detection across active research pipelines. Research interest: *empirical methods for detecting and characterizing failure modes in multi-agent LLM systems -- specifically, what the gap between a working demo and a reliable deployed system actually looks like at the retrieval and confidence-scoring layers.* M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

CORE COMPETENCIES

Large Language Model Evaluation & LLM-as-Judge | Empirical Research Methods | Agentic AI & Multi-Agent Systems (LangGraph) | Machine Learning (PyTorch, scikit-learn) | RAG Pipelines & Vector DBs | CI-Integrated Eval & Benchmarking Pipelines | Statistical Modeling & Hypothesis Testing | Python (Expert) | Production-Scale Systems | Open-Source & Open Source Development

EXPERIENCE

George Mason University

Aug 2024 - Aug 2025

Research Associate - College of Engineering and Computing · Virginia

- Achieved **$R^2 = 0.977$ regression accuracy** across 190+ countries by designing end-to-end research pipelines in Python and R on WHO and UK Road Safety datasets - data collection, transformation, hypothesis testing, and multi-source validation methodology driving faculty research strategy.
- Surfaced anomalies missed by manual review across 3 active research pipelines by applying **Isolation Forest** to automate data integrity checks - improved downstream model reliability and reduced false-negative rate in anomaly identification.
- Built research comprehension dashboards in **Tableau, ggplot2, and Quarto** for faculty leadership - improved research comprehension 40% across two cohorts through data-driven visualization of multi-variable findings.

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer Intern · Remote

- Built an agentic mission-planning system as a **LangGraph stateful graph** (parse → plan → retrieve → validate → conditional routing) - risk-aware compliance strategy grounding every decision in FAA Part 107 regulation; non-compliant missions route to rejection with cited reasons.
- Implemented a **RAG retrieval layer** over FAA Part 107 and operational-guidance corpus with cited-source answers - every compliance check functions as a hypothesis test grounded in source regulation; backend-agnostic retriever with documented ChromaDB/embedding upgrade path.
- Engineered validation guardrails enforcing altitude limits, battery budgets, and geometric no-fly-zone checks - blocked non-compliant missions before MAVLink/QGroundControl export across marketing and commercial survey operations.

George Mason University

Aug 2025 - May 2026

Graduate Teaching Assistant - MIS 303 · On-site · Part-time

- Mentored **600+ students** across multiple sections in Python programming (data structures, OOP, automation scripting) and database design (ER modeling, SQL, normalization).

Entrata - PropTech / Real Estate SaaS

Sep 2018 - Aug 2024

Senior Software Engineer · Remote

- Shipped 20+ full-stack production modules (Python, PHP, PostgreSQL, JavaScript) across 120+ Agile sprints for **100K+ monthly active users** - 99.5% uptime sustained across 6+ consecutive months.
- **Shipped Entrata's first production AI pipeline** across 1.6M real financial records (Python, PyTorch, OpenAI API) - custom neural network architecture and deep learning model for data transformation; automated workflows replacing manual Excel operations; output validation, quality gates, and compliance controls at launch.
- Achieved **100% PHI compliance** on AWS S3 migration with Libsodium-encrypted pipelines; optimized 200+ SQL queries (30% performance gain); eliminated 80% of manual operational workload via Python/Bash automation.
- Led microservices redesign using Docker - 50% improvement in system scalability, 30% reduction in deployment time; conducted 300+ peer code reviews and mentored 5+ junior engineers.

RESEARCH PROJECTS

Leasel - Agentic AI Platform for Affordable Housing Document Analysis & Ticket Investigation

Built a **LangGraph orchestration system** (Triage → Configuration → Historical Incident → Structured Data → Reasoning → Validation → Escalation nodes) with shared TypedDict state across all nodes; FastAPI service for document uploads and ticket submissions; SQLite storage written to swap to PostgreSQL with minimal friction. Retrieval is **TF-IDF cosine similarity** -- a deliberate starting point, not the final design: hybrid dense+BM25 with reranking is documented as the next step. Confidence scores are hand-tuned constants (0.8 for a matched setting, 0.6 for a similar ticket) rather than computed from retrieval signals -- the clearest remaining gap. Live **HUD User API** integration (Bearer-token auth, correct endpoint shapes) for real-time income-limit lookups on eligibility tickets. Validation node strips evidence-free claims and flags them unsupported rather than presenting them at face value; escalation packet fires when top-ranked cause confidence falls below threshold. **Read-only safety boundary hard-coded and test-verified**: the system never auto-resolves eligibility, configuration, or calculation decisions for tenants regardless of confidence score -- the deterministic compliance engine and human staff remain the source of truth, enforced in code not just documented. Validated end-to-end: correctly diagnosed a misconfigured property setting as the root cause of an incorrect ineligibility result -- without needing the hybrid retrieval layer still on the roadmap.

Python · LangGraph · FastAPI · TF-IDF · HUD User API · SQLite | github.com/venkatalakshmiparimala/Lease_complaince_rag

CrashClarity: Large-Scale Road Accident Analysis (104K Records)

Achieved **85%+ accuracy** on 104K+ UK accident records by building a Random Forest + Logistic Regression machine learning pipeline with feature engineering across weather, road, and time variables - hypothesis-driven analysis with train/test splits, cross-validation, and feature importance benchmarking; deployed interactive GitHub Pages dashboard.

Python · R · scikit-learn · Random Forest · Logistic Regression · Feature Engineering | Project Dashboard

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026

GPA: 3.87 | Focus: statistical modeling, ML systems, data pipelines, applied AI

B.Tech Computer Science & Engineering - JNTU Anantapur — 2017

TECHNICAL SKILLS

Research & Evaluation: Large Language Model Evaluation, LLM-as-Judge, Pass@K Harness, Empirical Methods, Hypothesis Testing, Statistical Modeling, Benchmark Design, Isolation Forest, Regression Analysis, Anomaly Detection

AI & Agents: LangChain, LangGraph, Multi-Agent Orchestration, RAG Pipelines, ChromaDB, Vector Database, Prompt Engineering, Anthropic API, OpenAI API, Groq, Ollama, FastAPI

ML & Data: PyTorch, Scikit-learn, XGBoost, Random Forest, Deep Learning, Neural Networks, Feature Engineering, NLP, BERT, Hugging Face, spaCy, Pandas, NumPy, Tableau, ggplot2, Quarto

Languages: Python, R, SQL, JavaScript, TypeScript, PHP, Bash

Infrastructure: Docker, AWS (S3, EC2, RDS), CI/CD (GitHub Actions), PostgreSQL, MySQL